COMPREHENSIVE ASSESSMENT OF MICROBIAL CONTAMINATION AND PHYSICOCHEMICAL CHARACTERISTICS OF RETAIL AND MODEL-FORMULATED DELI SALADS

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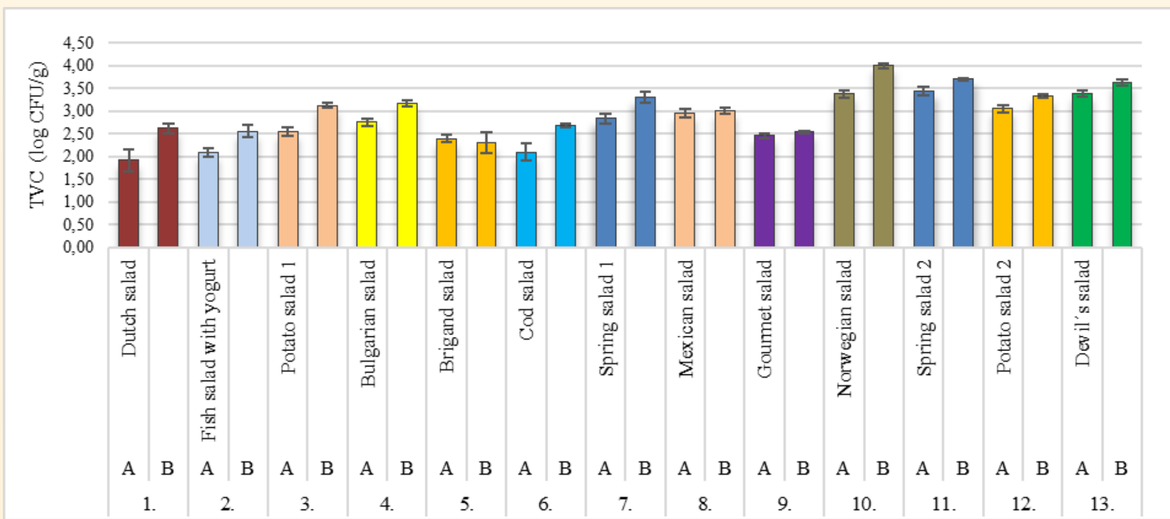
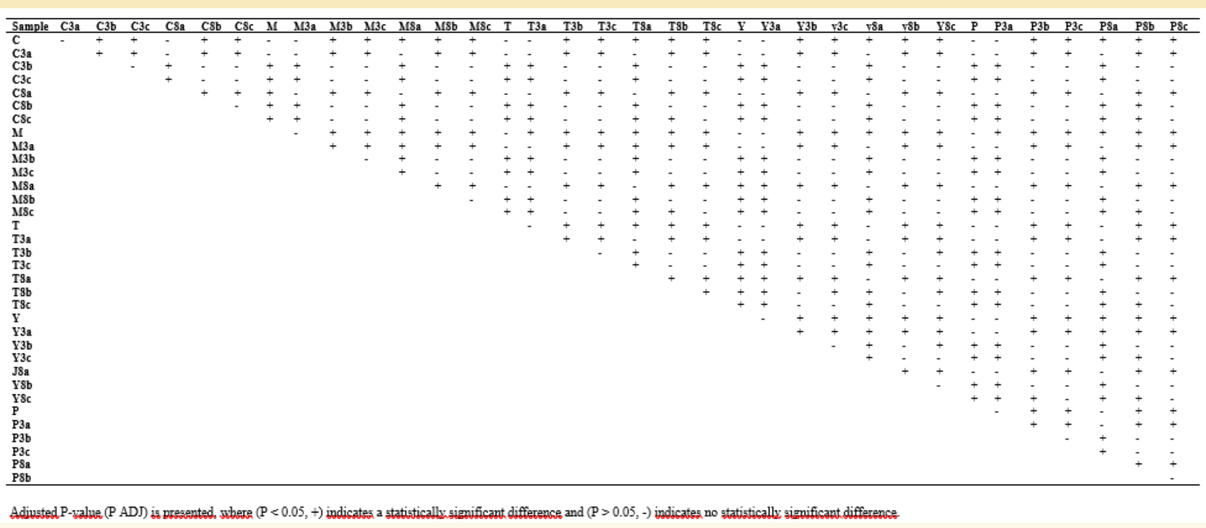
This study evaluated microbiological safety and selected physicochemical parameters of retail and model-formulated deli salads.

The dynamics of microbial contamination (total viable count – TVC, coliform bacteria, *Bacillus cereus*, microscopic filamentous fungi – MFF and yeasts) and physicochemical changes (pH, water activity, salt content) were monitored depending on sample type and storage conditions.

Thirteen commercial deli salads were purchased from retail display counters and analyzed immediately after purchase and after 24 h of storage at 2 °C.



Among retail samples, the highest TVC was detected in Norwegian salad (3.38 log CFU/g) and "Devil's" salad (3.37 log CFU/g), while the lowest was observed in Dutch salad (1.91 log CFU/g). Storage did not significantly affect TVC (P > 0.05). Coliform bacteria were detected in only one sample and remained within legislative limits. Microscopic filamentous fungi and yeasts occurred at low levels (<4×10¹ CFU/g). pH remained stable, whereas two samples exceeded the permitted salt content.



Sample	C3a	C3b	C3c	C8a	C8b	C8c	M	M3a	M3b	M3c	M8a	M8b	M8c	T	T3a	T3b	T3c	T8a	T8b	T8c	Y	Y3a	Y3b	y3c	y8a	y8b	Y8c	P	P3a	P3b	P3c	P8a	P8b	P8c
C	-	+	+	-	+	+	-	-	+	+	+	+	+	-	-	+	+	+	+	+	-	-	+	+	+	+	+	-	-	+	+	+	+	+
C3a		+	+	-	+	+	-	-	+	+	-	+	+	-	-	+	+	-	+	+	-	-	+	-	+	+	-	-	-	+	+	-	+	+
C3b			-	+	-	-	+	+	-	-	+	-	-	+	+	-	-	+	-	-	+	+	-	-	+	-	-	+	+	-	-	+	-	-
C3c				+	-	-	+	+	-	-	+	-	-	+	+	-	-	+	-	-	+	+	-	-	+	-	-	+	+	-	-	+	-	-
C8a					+	+	+	-	+	+	-	+	+	-	-	+	+	-	+	+	-	-	+	+	-	+	+	-	-	+	+	-	+	+
C8b						-	+	+	-	-	+	-	-	+	+	-	-	+	-	-	+	+	-	-	+	-	-	+	+	-	-	+	+	-
C8c							+	+	-	-	+	-	-	+	+	-	-	+	-	-	+	+	-	-	+	-	-	+	+	-	-	+	+	-
M								-	+	+	+	+	+	-	+	+	+	+	+	+	-	-	+	+	+	+	+	-	+	+	+	+	+	+
M3a									+	+	+	+	+	-	-	+	+	+	+	+	-	-	+	+	-	+	+	-	-	+	+	-	+	+
M3b										-	+	-	-	+	+	-	-	+	-	-	+	+	-	-	+	-	-	+	+	-	-	+	-	-
M3c											+	-	-	+	+	-	-	+	-	-	+	+	-	-	+	-	-	+	+	-	-	+	-	-
M8a												+	+	-	-	+	+	-	+	+	+	+	+	+	-	+	+	-	-	+	+	-	+	+
M8b													-	+	+	-	-	+	-	-	+	+	-	-	+	-	-	+	+	-	-	+	-	-
M8c														+	+	-	-	+	+	-	+	+	-	-	+	-	-	+	+	-	-	+	+	-
T															-	+	+	+	+	+	-	-	+	+	-	+	+	-	-	+	+	-	+	+
T3a																+	+	-	+	+	-	-	+	+	-	+	+	-	-	+	+	-	+	+
T3b																	-	+	-	-	+	+	-	-	+	-	-	+	+	-	-	+	-	-
T3c																		+	-	-	+	+	-	-	+	-	-	+	+	-	-	+	-	-
T8a																			+	+	+	+	+	+	-	+	+	-	-	+	+	-	+	+
T8b																				+	+	+	-	+	+	-	+	+	+	-	-	+	-	-
T8c																					+	+	-	-	+	-	-	+	+	-	-	+	+	-
Y																						-	+	+	+	+	+	-	-	+	+	+	+	+
Y3a																							+	+	+	+	+	-	-	+	+	+	+	+
Y3b																								-	+	-	+	+	+	-	-	+	-	-
Y3c																									+	-	-	+	+	-	-	+	+	-
J8a																										+	+	-	-	+	+	-	+	+
Y8b																											-	+	+	-	-	+	-	-
Y8c																												+	+	+	-	+	+	-
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P3a																														+	+	-	+	+
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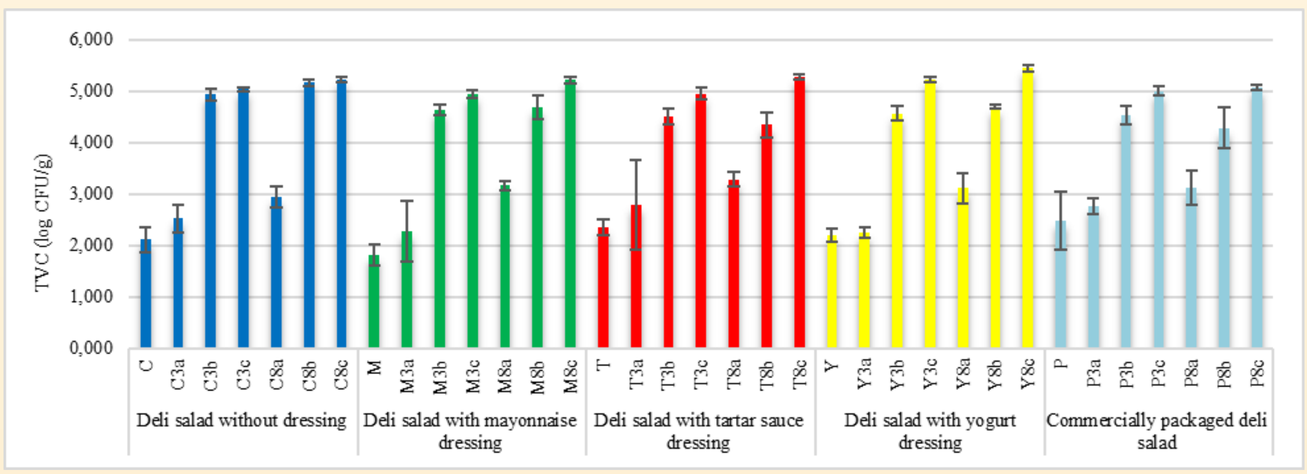
P3c																																+	-	-
P8a																																	+	+
P8b																																		-

Adjusted P-value (P ADJ) is presented, where (P < 0.05, +) indicates a statistically significant difference and (P > 0.05, -) indicates no statistically significant difference.

The second part of the study included one commercial deli salad and four laboratory-prepared model salads simulating gastronomic practice. All model ready-to-eat salads (*Vlašský šalát*) contained identical ingredients but differed in dressing (no dressing, mayonnaise, tartar sauce, yoghurt). Analyses were performed immediately after preparation and after 12, 24 and 48 h of storage at 3 °C and 8 °C.



In model salads, storage temperature and time significantly influenced microbial growth (P < 0.05), with faster proliferation at 8 °C. Dressings showed an initial inhibitory effect which decreased after 48 h. After 48 h at 8 °C, the lowest TVC was observed in mayonnaise salad (5.22 log CFU/g) and the highest in yoghurt salad (5.45 log CFU/g). Coliform bacteria were absent; however, yoghurt salad exceeded the limit for *Bacillus cereus* (4.13 log CFU/g). Yeasts and filamentous fungi remained low in all samples. Storage caused slight increases in pH and water activity. All model salads prepared according to commonly available recipes exceeded the maximum salt content, except the yoghurt variant.

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